In class lab #7 for Numerical Analysis

Your name: _____

Submit both your code and results before 23:00 Thur, Apr 18 or you will get 0.

Part I: Least Squares and the Normal Equations (45 points)

Problem 1. (15 points)

(a) Note that for introducing the idea of Least Squares, we said that the Least Squares comprise the two main steps in the lecture, that is,

1. formulate the error function in a square sense;
2. Determine unknowns by minimize this error function.

Please use these two steps to find a straight line $p_1(x) = a_0 + a_1 x$ through the set of data using least squares:

x	$x_0 = 0$	$x_1 = 1$	$x_2 = 2$	$x_3 = 5$
y	$y_0 = 0$	$y_1 = 3$	$y_2 = 3$	$y_3 = 6$

(b) Note that for the above problem, we solve a 2×2 linear system instead of solving the following overdetermined system

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 3 \\ 6 \end{bmatrix}, \tag{1}$$

which can be denoted as $A\mathbf{a} = \mathbf{b}$, where $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix}$, $\mathbf{a} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} 0 \\ 3 \\ 3 \\ 6 \end{bmatrix}$.

Type the following lines into the command windows of MATLAB, and present the result.

```
>> x = [0 , 1 , 2 , 5]'

x =
```

```
>> y = [0 , 3 , 3 , 6]'

y =
```

```
>> A=[x x.^0]

A =
```

```
>> c=A'*A \ A'*y

c =
```

Problem 2. (15 points) What is the "slope" of the parabola $y = x^2$ on $[0, 1]$? Find the best least squares line that fits the parabola at n evenly spaced points in the interval.

(a) $n = 10$, plot the parabola and the lines.

(b) $n = 20$, Plot the parabola and the lines.

(c) What do you expect the result to be as $n \to \infty$?

Problem 3. (15 points) Let A be the $10 \times n$ matrix formed by the first n columns of the 10×10 Hilbert matrix. Let c be the n-vector $[1, \cdots, 1]$, and set $b = Ac$. (Hint: use MATLAB code hilb(n) to generate an $n \times n$ Hilbert matrix)

(a) Use the normal equations to solve the least squares problem $Ax = b$ for $n = 6$, and compare with the correct least squares solution $x = c$. How many correct decimal places can be computed?

(b) Use the normal equations to solve the least squares problem $Ax = b$ for $n = 8$, and compare with the correct least squares solution $x = c$. How many correct decimal places can be computed?

(c) Use condition number to explain the results in (a) and (b).

Part II: Newton's Divided Differences (55 points)

Creat a Newton.m to find a polynomial of degree 3 $P_3(x)$ that passes through the points $(0, 1), (2, 3), (3, 0), (4, 2)$ and evaluate the value $P_3(2.5)$. Write down the code and its result.